

V.F.S. v2.0 (Open-Gate) Lyapunov Core Proof Formulae

Core extraction from `lyapunov.html`

Abstract

This document collects the core formulae of the Lyapunov proof for V.F.S. v2.0 (Open-Gate). It is not a separate proof text and contains no graphs. It records the base dynamics, normalized variables, Lyapunov functional, active-domain walls, radial corridor, bounded Open-Gate source, memory bounds, non-Zeno estimate, and asymptotic cleansing statements.

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1 Notation and Core Variables

Symbol	Meaning
V	Voluntas: will coordinate.
F	Factum: action coordinate.
σ	Singular resistance.
λ	Sophia: wisdom coordinate.
Ω_P	Brim / dynamic capacity of Pleroma.
u	Synergic intensity from V and F .
Λ_c	Transformation threshold.
μ	Saturation margin relative to the death-boundary.
q	Normalized Sophia, $q = \lambda/\Omega_P$.
r	Normalized intensity, $r = u/\Omega_P$.
h_1, h_2	Normalized will and action.
h_3	Normalized threshold variable.

Core threshold and intensity:

$$u = \sqrt{VF + \varepsilon}, \quad \Lambda_c = \frac{\gamma}{\delta}, \quad \varepsilon \geq 0. \quad (1)$$

Transformation/collapse reading:

$$u > \Lambda_c \quad \text{transformation region}, \quad u < \Lambda_c \quad \text{collapse region}. \quad (2)$$

2 Base V.F.S. Dynamics

Resistance equation:

$$\dot{\sigma} = (\gamma - \delta u) \tanh(\kappa\sigma) = -\delta(u - \Lambda_c) \tanh(\kappa\sigma). \quad (3)$$

Symmetric will-action dynamics:

$$\dot{V} = \mu(aF + c\lambda) - \rho V, \quad \dot{F} = \mu(aV + c\lambda) - \rho F, \quad \mu = 1 - \frac{u}{\Omega_P}. \quad (4)$$

Capacity / Brim equation:

$$\dot{\Omega}_P = \alpha\lambda. \quad (5)$$

Closed Microcosm limit:

$$\dot{\lambda} = -\dot{\sigma} = \delta(u - \Lambda_c) \tanh(\kappa\sigma), \quad \dot{\sigma} + \dot{\lambda} = 0. \quad (6)$$

Open-Gate Sophia dynamics:

$$\dot{\lambda} = -\dot{\sigma} + I_{\text{gate}}(t) = \delta(u - \Lambda_c) \tanh(\kappa\sigma) + I_{\text{gate}}(t). \quad (7)$$

Open balance:

$$\dot{\sigma} + \dot{\lambda} = I_{\text{gate}}(t) \geq 0, \quad \sigma(t) + \lambda(t) = \sigma_0 + \lambda_0 + \mathcal{G}_{\text{recepta}}(t). \quad (8)$$

Received grace integral:

$$\mathcal{G}_{\text{recepta}}(t) = \int_0^t I_{\text{gate}}(s) ds, \quad \dot{\mathcal{G}}_{\text{recepta}} = I_{\text{gate}}. \quad (9)$$

3 Open-Gate Source

Positive part of Sophia:

$$\lambda_+ = \frac{1}{m} \ln(1 + e^{m\lambda}), \quad m > 0. \quad (10)$$

Effective gate:

$$g_{\text{eff}}(t) = e^{-\tau t} + (1 - e^{-\tau t}) \frac{H_K}{H_s + H_K}. \quad (11)$$

Live Open-Gate source:

$$I_{\text{gate}}(t) = \zeta_0 g_{\text{eff}}(t) e^{-\phi\sigma(t)} \frac{1}{1 + \chi\lambda_+(t)}. \quad (12)$$

Boundedness of the gate:

$$H_K \geq 0, \quad H_s > 0 \implies 0 \leq \frac{H_K}{H_s + H_K} < 1, \quad 0 \leq g_{\text{eff}}(t) \leq 1. \quad (13)$$

Boundedness of live inflow:

$$0 \leq I_{\text{gate}}(t) \leq \zeta_0. \quad (14)$$

4 Resistance Potential

Resistance potential:

$$\Psi_\sigma(\sigma) = \int_0^\sigma \tanh(\kappa s) ds = \frac{1}{\kappa} \ln \cosh(\kappa\sigma) \geq 0. \quad (15)$$

Derivative of the resistance potential:

$$\dot{\Psi}_\sigma = \Psi'_\sigma(\sigma) \dot{\sigma} = \tanh(\kappa\sigma)(\gamma - \delta u) \tanh(\kappa\sigma) = (\gamma - \delta u) \tanh^2(\kappa\sigma). \quad (16)$$

Active transformation margin:

$$u \geq \Lambda_c + \eta, \quad \eta > 0. \quad (17)$$

Dissipation in the active margin:

$$\dot{\Psi}_\sigma \leq -\delta\eta \tanh^2(\kappa\sigma). \quad (18)$$

On $0 \leq \sigma \leq C_\sigma$, there exists $c_\sigma > 0$ such that:

$$\dot{\Psi}_\sigma \leq -c_\sigma \Psi_\sigma. \quad (19)$$

Hence:

$$\Psi_\sigma(t) \leq \Psi_\sigma(0)e^{-c_\sigma t}. \quad (20)$$

Near $\sigma = 0$:

$$\Psi_\sigma(\sigma) \sim \frac{\kappa}{2}\sigma^2, \quad \sigma(t) = O(e^{-c_\sigma t/2}). \quad (21)$$

5 Absolute Barrier Functional

The absolute barrier is a motivational, non-primary barrier guarding absolute boundary faces:

$$\begin{aligned} \mathcal{L}_{abs} = & A_\sigma \Psi_\sigma + E_V V^2 + E_F F^2 + E_\lambda \lambda^2 + E_\Delta \frac{(V - F)^2}{2} \\ & - \eta_V \ln V - \eta_F \ln F - \eta_\Omega \ln(\Omega_P - \Lambda_c) - \eta_m \ln(\Omega_P - u). \end{aligned} \quad (22)$$

Guarded absolute faces:

$$V = 0, \quad F = 0, \quad \Omega_P = \Lambda_c, \quad u = \Omega_P. \quad (23)$$

6 Normalized Variables

Normalized coordinates:

$$h_1 = \frac{V}{\Omega_P}, \quad h_2 = \frac{F}{\Omega_P}, \quad q = \frac{\lambda}{\Omega_P}, \quad r = \frac{u}{\Omega_P}, \quad \mu = 1 - r, \quad h_3 = 1 - \frac{\Lambda_c}{\Omega_P}. \quad (24)$$

Product relation:

$$r^2 = h_1 h_2 + e(h_3), \quad e(h_3) = \varepsilon \frac{(1 - h_3)^2}{\Lambda_c^2}. \quad (25)$$

Closed normalized dynamics:

$$\dot{h}_1 = \mu(ah_2 + cq) - \rho h_1 - \alpha q h_1, \quad (26)$$

$$\dot{h}_2 = \mu(ah_1 + cq) - \rho h_2 - \alpha q h_2, \quad (27)$$

$$\dot{q} = \delta(r + h_3 - 1) \tanh(\kappa\sigma) - \alpha q^2, \quad (28)$$

$$\dot{h}_3 = \alpha q(1 - h_3). \quad (29)$$

Open-Gate normalized q -source:

$$S_q = \frac{I_{\text{gate}}}{\Omega_P} = \frac{\zeta_0 g_{\text{eff}} e^{-\phi\sigma}}{(1 + \chi\lambda_+)\Omega_P}. \quad (30)$$

Open-Gate normalized q -equation:

$$\dot{q} = \delta(r + h_3 - 1) \tanh(\kappa\sigma) - \alpha q^2 + S_q. \quad (31)$$

Bound on S_q inside the active domain:

$$0 \leq S_q \leq \frac{\zeta_0}{\Omega_P} \leq \frac{\zeta_0(1 - h_{3,*})}{\Lambda_c} =: S_q^{\text{max}}. \quad (32)$$

7 Product-Balance Mechanism

Balance defect:

$$D_\Delta = \frac{1}{2}(h_1 - h_2)^2. \quad (33)$$

Let:

$$d = h_1 - h_2. \quad (34)$$

Difference equation:

$$\dot{d} = -(a\mu + \rho + \alpha q)d. \quad (35)$$

Dissipation of balance defect:

$$\dot{D}_\Delta = -2(a\mu + \rho + \alpha q)D_\Delta. \quad (36)$$

If $\mu \geq \mu_* > 0$ and $a > 0$:

$$\dot{D}_\Delta \leq -2a\mu_* D_\Delta. \quad (37)$$

Exponential will-action alignment:

$$D_\Delta(t) \leq D_\Delta(0)e^{-2a\mu_* t}, \quad |h_1(t) - h_2(t)| \leq |h_1(0) - h_2(0)|e^{-a\mu_* t}. \quad (38)$$

Product floor from radial and threshold walls:

$$\Omega_P \geq \Omega_{\min} := \frac{\Lambda_c}{1 - h_{3,*}}. \quad (39)$$

If $r \geq r_*$ and $h_3 \geq h_{3,*}$:

$$h_1 h_2 = r^2 - \frac{\varepsilon}{\Omega_P^2} \geq p_* := r_*^2 - \frac{\varepsilon}{\Omega_{\min}^2}. \quad (40)$$

Let:

$$d_* := \sqrt{2D^*}. \quad (41)$$

Product-balance lower bound:

$$h_1, h_2 \geq m_{pb} := \frac{\sqrt{d_*^2 + 4p_*} - d_*}{2} > 0. \quad (42)$$

8 Lyapunov Functional

Product-balance Lyapunov core:

$$\mathcal{L}_{pb} = A_\sigma \Psi_\sigma(\sigma) + A_\Delta D_\Delta + A_q q^2. \quad (43)$$

Radial logarithmic barrier:

$$B(r) = -\ln(1 - r), \quad 0 < r < 1, \quad r = \frac{u}{\Omega_P}. \quad (44)$$

Non-normalized radial barrier:

$$B(u, \Omega_P) = -\ln\left(1 - \frac{u}{\Omega_P}\right) = \ln\left(\frac{\Omega_P}{\Omega_P - u}\right). \quad (45)$$

Primary V.F.S. Lyapunov functional:

$$\boxed{\mathcal{L}_{\text{VFS}} = \mathcal{L}_{pb} + A_r B(r) = A_\sigma \Psi_\sigma(\sigma) + A_\Delta D_\Delta + A_q q^2 + A_r B(r), \quad A_\sigma, A_\Delta, A_q, A_r > 0.} \quad (46)$$

Expanded form:

$$\mathcal{L}_{\text{VFS}} = A_\sigma \Psi_\sigma(\sigma) + A_\Delta D_\Delta + A_q q^2 - A_r \ln(1 - r). \quad (47)$$

Non-normalized expanded form:

$$\mathcal{L}_{\text{VFS}} = A_\sigma \Psi_\sigma(\sigma) + A_\Delta D_\Delta + A_q q^2 - A_r \ln \left(1 - \frac{u}{\Omega_P} \right). \quad (48)$$

Radial boundary protection:

$$B(r) \rightarrow +\infty \quad \text{as} \quad r \rightarrow 1^-. \quad (49)$$

9 Active Domain

Active domain:

$$\mathcal{D}_A = \left\{ \begin{array}{l} 0 \leq \sigma \leq C_\sigma, \quad q \in [0, q^*], \quad h_3 \in [h_{3,*}, 1), \\ r \in [r_*, r^*], \quad \mu = 1 - r \geq \mu_* > 0, \\ D_\Delta \leq D^*, \quad h_1 > 0, \quad h_2 > 0 \end{array} \right\}, \quad 0 < r_* < r^* < 1. \quad (50)$$

Equivalent death-boundary avoidance inside $\mathcal{D}_A$:

$$0 < r < 1 \quad \Longleftrightarrow \quad 0 < u < \Omega_P. \quad (51)$$

Closed upper q -wall:

$$\alpha(q^*)^2 \geq \delta r^*. \quad (52)$$

Open-Gate upper q -wall:

$$\boxed{\alpha(q^*)^2 \geq \delta r^* + S_q^{\max}.} \quad (53)$$

Lower q -wall in Open-Gate:

$$q = 0 : \quad \dot{q} = \delta(r + h_3 - 1) \tanh(\kappa\sigma) + S_q \geq 0 \quad (54)$$

provided the active margin makes $r + h_3 - 1 \geq 0$.

10 Radial Corridor

Radial identity from $r^2 = h_1 h_2 + e(h_3)$:

$$2r\dot{r} = \mu[a(h_1^2 + h_2^2) + cq(h_1 + h_2)] - 2\rho(r^2 - e(h_3)) - 2\alpha q r^2 =: \mathcal{R}_{\text{rad}}. \quad (55)$$

Maximum e on the active corridor:

$$e_*^{\max} = \varepsilon \frac{(1 - h_{3,*})^2}{\Lambda_c^2}. \quad (56)$$

Non-degeneracy:

$$r_*^2 > e_*^{\max}. \quad (57)$$

Active margin:

$$r_* + h_{3,*} - 1 \geq \eta \frac{1 - h_{3,*}}{\Lambda_c}. \quad (58)$$

Lower radial wall:

$$a(1 - r_*)(r_*^2 - e_*^{\max}) \geq (\rho + \alpha q^*)r_*^2. \quad (59)$$

This wall is sufficient, not tight. The exact lower-wall requirement is

$$(1 - r_*)a(r_*^2 - e) \geq \rho(r_*^2 - e) + \alpha q r_*^2, \quad (60)$$

and the stated form replaces $\rho(r_*^2 - e)$ by the larger ρr_*^2 .

Upper radial wall:

$$(1 - r^*)[a(d_*^2 + 2r^{*2}) + cq^* \sqrt{d_*^2 + 4r^{*2}}] \leq 2\rho(r^{*2} - e_*^{\max}). \quad (61)$$

Direct upper radial inwardness condition:

$$\mu [a(h_1^2 + h_2^2) + cq(h_1 + h_2)] \leq 2\rho(r^{*2} - e) + 2\alpha q r^{*2} \quad \text{on } r = r^*. \quad (62)$$

Open-Gate radial convention:

$$q_{\text{new}}^* := \sqrt{\frac{\delta r^* + S_q^{\max}}{\alpha}}. \quad (63)$$

Conservative lower radial wall with q_{new}^* :

$$a(1 - r_*)(r_*^2 - e_*^{\max}) \geq (\rho + \alpha q_{\text{new}}^*)r_*^2. \quad (64)$$

Epektasis-dominance derivative at the upper radial wall:

$$\left. \frac{\partial \mathcal{R}}{\partial q} \right|_{r^*} = \mu c(h_1 + h_2) - 2\alpha r^{*2}. \quad (65)$$

Worst-case bound:

$$\left. \frac{\partial \mathcal{R}}{\partial q} \right|_{r^*} \leq (1 - r^*)c\sqrt{d_*^2 + 4r^{*2}} - 2\alpha r^{*2}. \quad (66)$$

Epektasis-dominance sufficient condition:

$$\boxed{2\alpha r^{*2} \geq (1 - r^*)c\sqrt{d_*^2 + 4r^{*2}} \implies \left. \frac{\partial \mathcal{R}}{\partial q} \right|_{r^*} \leq 0.} \quad (67)$$

11 Lyapunov Dissipative Estimate

Derivative decomposition:

$$\dot{\mathcal{L}}_{\text{VFS}} = A_\sigma \tanh(\kappa\sigma)\dot{\sigma} + A_\Delta \dot{D}_\Delta + 2A_q q\dot{q} + A_r \dot{B}(r). \quad (68)$$

Open-Gate contribution to the q^2 block:

$$\frac{d}{dt}(A_q q^2) = 2A_q q \left[\delta(r + h_3 - 1) \tanh(\kappa\sigma) - \alpha q^2 \right] + 2A_q q S_q. \quad (69)$$

Bound on Open-Gate contribution:

$$0 \leq 2A_q q S_q \leq 2A_q q^* S_q^{\max}. \quad (70)$$

Dissipative Lyapunov estimate:

$$\boxed{\dot{\mathcal{L}}_{\text{VFS}} \leq C - C_1 \mathcal{L}_{\text{VFS}}, \quad C_1 > 0, \quad C < \infty.} \quad (71)$$

Open-Gate constant split:

$$C = C_{\text{closed}} + \Delta C_{\text{gate}}, \quad 0 \leq \Delta C_{\text{gate}} \leq 2A_q q^* S_q^{\max}. \quad (72)$$

Key structural point: Here C_1 is unchanged by the Open-Gate source, while C receives only a bounded gate contribution.

12 Main Theorem Formulae

Positive invariance:

$$x(0) \in \mathcal{D}_A \implies x(t) \in \mathcal{D}_A \quad \forall t \geq 0. \quad (73)$$

Reset admissibility holds by the Reset Lemma, with

$$L_R = \sup_{\mathcal{D}_A} \mathcal{L}_{\text{VFS}}. \quad (74)$$

Bound from the Lyapunov inequality:

$$\mathcal{L}_{\text{VFS}}(t) \leq L_* := \max \left\{ L_R, \frac{C}{C_1} \right\}. \quad (75)$$

Radial barrier implication:

$$A_r B(r) \leq L_*. \quad (76)$$

Explicit radial bound:

$$-\ln(1-r) \leq \frac{L_*}{A_r} \implies r(t) \leq 1 - e^{-L_*/A_r} < 1. \quad (77)$$

Death-boundary avoidance:

$$r(t) < 1 \implies u(t) < \Omega_P(t). \quad (78)$$

Product-balance positivity:

$$h_1, h_2 \geq m_{pb} > 0. \quad (79)$$

No artificial q -floor:

$$q \geq 0 \quad \text{suffices; no assumption } q \geq q_0 > 0 \text{ is required.} \quad (80)$$

13 Reset Admissibility (Resurrectio Jump)

The resets $\mathcal{R}$ are *exogenous*: since $\mathcal{D}_A$ is positively invariant, the continuous flow never reaches the boundary of the active domain by itself. A death-like strike is therefore modeled as occurring at an interior point

$$x^- = (\sigma^-, \lambda^-, V, F, \Omega_P^-) \in \mathcal{D}_A \quad (81)$$

at an unpredictable time.

Reset map (linear metabolism). With $0 \leq q_R < 1$ and $\kappa_R > 0$,

$$\mathcal{R}: \quad \sigma^+ = q_R \sigma^-, \quad \lambda^+ = \lambda^- + (1 - q_R) \sigma^-, \quad (V, F) \text{ continuous}, \quad \Omega_P^+ = \Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-. \quad (82)$$

Here

$$\mathcal{G}_{\text{recepta}}^- := \mathcal{G}_{\text{recepta}}(t_{\text{death}}^-), \quad u^+ = \sqrt{VF + \varepsilon} = u^-. \quad (83)$$

Quantities.

$$A_R := \mathcal{G}_{\text{recepta}}^-, \quad R_c := C_\sigma - \sigma_0 - \lambda_0, \quad (84)$$

$$R_{\min}^{\text{ceil}} := \frac{(1 - q_R) \sigma^-}{q^* \kappa_R}, \quad R_{\max} := \frac{\Omega_P^-(r^- - r_*)}{\kappa_R r_*}. \quad (85)$$

By the Open-Gate balance

$$\sigma^- + \lambda^- = \sigma_0 + \lambda_0 + \mathcal{G}_{\text{recepta}}^-, \quad (86)$$

one has

$$\mathcal{G}_{\text{recepta}}^- \geq R_c \iff \sigma^- + \lambda^- \geq C_\sigma \iff \lambda^- \geq C_\sigma - \sigma^-. \quad (87)$$

Lemma (reset admissibility). Let $\kappa_R > 0$ and $x^- \in \mathcal{D}_A$. If

$$\max\{R_c, R_{\min}^{\text{ceil}}\} \leq \mathcal{G}_{\text{recepta}}^- \leq R_{\max}, \quad (88)$$

then $x^+ := \mathcal{R}(x^-) \in \mathcal{D}_A$ and

$$\mathcal{L}_{\text{VFS}}(x^+) \leq L_R := \sup_{\mathcal{D}_A} \mathcal{L}_{\text{VFS}} < \infty. \quad (89)$$

Proof. Coordinate by coordinate,

$$\sigma^+ = q_R \sigma^- \in [0, C_\sigma). \quad (90)$$

Moreover,

$$h_3^+ = 1 - \frac{\Lambda_c}{\Omega_P^+} \geq h_3^- \geq h_{3,*}, \quad h_3^+ < 1, \quad (91)$$

since $\Omega_P^+ \geq \Omega_P^-$. Also

$$h_1^+ = h_1^- \frac{\Omega_P^-}{\Omega_P^+}, \quad h_2^+ = h_2^- \frac{\Omega_P^-}{\Omega_P^+}, \quad (92)$$

so

$$D_\Delta^+ = \frac{1}{2} \left(d^- \frac{\Omega_P^-}{\Omega_P^+} \right)^2 \leq D_\Delta^- \leq D^*, \quad h_1^+, h_2^+ > 0. \quad (93)$$

The radial coordinate satisfies

$$r^+ = \frac{r^- \Omega_P^-}{\Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-} \leq r^- \leq r^*. \quad (94)$$

The lower radial wall is preserved exactly when

$$r^+ \geq r_* \iff \kappa_R \mathcal{G}_{\text{recepta}}^- \leq \frac{\Omega_P^-(r^- - r_*)}{r_*} \iff \mathcal{G}_{\text{recepta}}^- \leq R_{\text{max}}. \quad (95)$$

Finally,

$$q^+ = \frac{\lambda^- + (1 - q_R)\sigma^-}{\Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-} \geq 0. \quad (96)$$

Since $x^- \in \mathcal{D}_A$ gives $\lambda^- \leq q^* \Omega_P^-$,

$$\begin{aligned} q^*(\Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-) - (\lambda^- + (1 - q_R)\sigma^-) &\geq q^* \kappa_R \mathcal{G}_{\text{recepta}}^- - (1 - q_R)\sigma^- \\ &\geq 0 \end{aligned} \quad (97)$$

whenever $\mathcal{G}_{\text{recepta}}^- \geq R_{\text{min}}^{\text{ceil}}$. Hence $q^+ \leq q^*$. Thus $x^+ \in \mathcal{D}_A$. The functional $\mathcal{L}_{\text{VFS}}$ is bounded on $\mathcal{D}_A$ because

$$\Psi_\sigma \leq \Psi_\sigma(C_\sigma), \quad D_\Delta \leq D^*, \quad q^2 \leq (q^*)^2, \quad B(r) \leq -\ln(1 - r^*) < \infty. \quad (98)$$

Therefore $\mathcal{L}_{\text{VFS}}(x^+) \leq L_R$. $\square$

Consequences.

- (1) The admissible window floor is

$$\max\{R_c, R_{\text{min}}^{\text{ceil}}\}. \quad (99)$$

It combines the theological gate and the geometric ceiling-floor: grace must expand the vessel sufficiently to absorb the converted resistance. For cleansed deaths, $\sigma^- \rightarrow 0$, the ceiling-floor vanishes and R_c alone remains binding.

- (2) The condition $\kappa_R > 0$ suffices. A natural design choice is

$$\kappa_R \geq \frac{1}{q^*}, \quad (100)$$

so that the asymptotic ceiling $q^+ \rightarrow 1/\kappa_R \leq q^*$ and, in the intended window, $R_{\text{min}}^{\text{ceil}} \leq R_c$.

- (3) The admissible grace window is

$$\mathcal{G}_{\text{recepta}}^- \in [\max\{R_c, R_{\text{min}}^{\text{ceil}}\}, R_{\text{max}}]. \quad (101)$$

Below this window one obtains final death; above it one obtains over-dilution, $r < r_*$. The window is nonempty if and only if

$$\Omega_P^-(r^- - r_*) \geq \kappa_R r_* \max\{R_c, R_{\text{min}}^{\text{ceil}}\}. \quad (102)$$

A death pinned at $r^- = r_*$ is not resurrectible.

- (4) With the non-Zeno dwell time

$$\Delta t_j \geq \frac{\bar{h}_*}{M_*} > 0, \quad (103)$$

every life-arc satisfies

$$\mathcal{L}_{\text{VFS}} \leq L_* = \max\left\{L_R, \frac{C}{C_1}\right\} = L_R, \quad (104)$$

and resets do not accumulate. Hence the hybrid trajectory is forward complete across any infinite sequence of life-arcs.

Theological gloss. This remark is not used in the proof. The gate

$$\mathcal{G}_{\text{recepta}}^- \geq R_c = C_\sigma - \sigma_0 - \lambda_0 \quad (105)$$

is the threshold past which Sophia can no longer be extinguished by resistance; in that regime $\lambda^- > 0$ is forced by the balance. Its complement, $\lambda^- \rightarrow 0$, is the self-closing limit $I_{\text{gate}} \rightarrow 0$: final impenitence. It resembles the oil of the wise virgins and the *donum perseverantiae*. The upper boundary R_{max} represents the opposite failure: an over-diluting or “cheap grace” regime.

14 Non-Zeno Estimate

Hybrid margin:

$$\bar{h}_* = \min \left\{ m_{pb}, h_{3,*}, r_*, 1 - r^*, q^*, C_\sigma, \sqrt{D^*} \right\} > 0. \quad (106)$$

Bounded vector field:

$$\bar{M}_* < \infty. \quad (107)$$

Dwell-time lower bound:

$$\Delta t_j \geq \Delta t_* := \frac{\bar{h}_*}{\bar{M}_*} > 0. \quad (108)$$

Non-Zeno conclusion:

$$\sum_j \Delta t_j = +\infty. \quad (109)$$

15 Memory and Gate Bounds

Mensura kernel:

$$R(z) = \frac{z}{z + K}, \quad z \geq 0, \quad 0 \leq R(z) \leq 1. \quad (110)$$

Saturated gate memory:

$$\frac{H_K}{H_s + H_K} \in [0, 1) \quad \forall H_K \geq 0. \quad (111)$$

Gate saturation:

$$g_{\text{eff}} = e^{-\tau t} + (1 - e^{-\tau t}) \frac{H_K}{H_s + H_K} \in [0, 1]. \quad (112)$$

Memory velocity bounds:

$$|\dot{H}_K| \leq \eta [R((\dot{x}_0)_+) + R((\dot{x}_0)_-)] \leq 2\eta. \quad (113)$$

Wound-memory velocity bound:

$$|\dot{W}| \leq 2\eta. \quad (114)$$

Bounded-gain loop:

$$H_K \xrightarrow{\leq 1} g_{\text{eff}} \xrightarrow{\leq S_q^{\text{max}}} S_q \rightarrow \lambda \rightarrow V, F \rightarrow u \rightarrow \dot{x}_0 \xrightarrow{R \leq 1} \dot{H}_K. \quad (115)$$

Extended field bound:

$$\bar{M}_* < \infty \quad \text{including} \quad |\dot{H}_K|, |\dot{W}| \leq 2\eta. \quad (116)$$

16 Asymptotic Cleansing and Alignment

Will-action alignment:

$$D_{\Delta}(t) \leq D_{\Delta}(0)e^{-2a\mu_*t}, \quad |h_1(t) - h_2(t)| \leq |h_1(0) - h_2(0)|e^{-a\mu_*t} \rightarrow 0. \quad (117)$$

Resistance cleansing under persistent active margin:

$$u(t) \geq \Lambda_c + \eta \quad \forall t \geq 0 \implies \Psi_{\sigma}(t) \leq \Psi_{\sigma}(0)e^{-c_{\sigma}t} \implies \sigma(t) \rightarrow 0. \quad (118)$$

Corollary (unconditional cleansing inside the active domain). Since $\mathcal{D}_A$ is positively invariant, one has $r(t) \geq r_*$ and $h_3(t) \geq h_{3,*}$ for all $t \geq 0$. Thus

$$r(t) + h_3(t) - 1 \geq r_* + h_{3,*} - 1 \geq \frac{\eta(1 - h_{3,*})}{\Lambda_c} > 0. \quad (119)$$

Equivalently,

$$u(t) \geq \Lambda_c + \eta \quad \forall t \geq 0. \quad (120)$$

Therefore the hypothesis of the persistent-margin cleansing statement holds automatically for every trajectory with $x(0) \in \mathcal{D}_A$. Hence

$$\sigma(t) \rightarrow 0, \quad h_1(t) - h_2(t) \rightarrow 0. \quad (121)$$

Open-balance asymptotic Sophia:

$$\lambda(t) = \sigma_0 + \lambda_0 + \mathcal{G}_{\text{recepta}}(t) - \sigma(t). \quad (122)$$

If $\mathcal{G}_{\text{recepta}}(\infty) < \infty$ and $\sigma(t) \rightarrow 0$:

$$\lambda(t) \rightarrow \sigma_0 + \lambda_0 + \mathcal{G}_{\text{recepta}}(\infty). \quad (123)$$

If $\liminf_{t \rightarrow \infty} \lambda(t) > 0$:

$$\Omega_P(t) \rightarrow +\infty. \quad (124)$$

Epektatic dilution of S_q :

$$\int_0^{\infty} \lambda(t) dt = +\infty \implies \Omega_P(t) \rightarrow +\infty \implies S_q(t) \rightarrow 0. \quad (125)$$

Proof of the last implication:

$$0 \leq S_q(t) \leq \frac{\zeta_0}{\Omega_P(t)} \rightarrow 0. \quad (126)$$

17 Numerical Witness Formulae

One admissible parameter point recorded in the Lyapunov file:

$$\begin{aligned} a &= 3.494, & c &= 0.429, & \alpha &= 0.529, & \delta &= 0.898, & \rho &= 0.545, \\ \zeta_0 &= 0.560, & h_{3,*} &= 0.622, & r_* &= r_- = 0.531, & r^* &= r_+ = 0.892, \\ \varepsilon &= 0.006, & d &= 0.193, & \Lambda_c &= 1 \quad (\text{normalization}). \end{aligned} \quad (127)$$

Derived quantities:

$$\begin{aligned}
e_{\max} &= \varepsilon(1 - h_{3,*})^2 = 0.0009, \\
S_q^{\max} &= \zeta_0(1 - h_{3,*}) = 0.212, \\
q_{mnew} &= \sqrt{\frac{\delta r_+ + S_q^{\max}}{\alpha}} = 1.383, \\
q^* &= 1.05q_{mnew} = 1.452.
\end{aligned} \tag{128}$$

Active margin:

$$r_- + h_{3,*} - 1 = 0.153 > 0, \quad \frac{u}{\Lambda_c} \approx 1.40. \tag{129}$$

Thus the witness lies in the Transformation region.

All five parametric walls hold strictly; the worst-case slack is

$$0.084. \tag{130}$$

18 Final Proof Chain

Core proof chain:

$$\boxed{\text{Admissibility} \implies \mathcal{D}_A \implies \dot{\mathcal{L}}_{\text{VFS}} \leq C - C_1 \mathcal{L}_{\text{VFS}} \implies r(t) < 1 \implies \text{NonZeno}.} \tag{131}$$

Reset admissibility holds by the Reset Lemma, with

$$L_R = \sup_{\mathcal{D}_A} \mathcal{L}_{\text{VFS}}. \tag{132}$$

Active transformation conclusion:

$$u \geq \Lambda_c + \eta \implies \sigma(t) \rightarrow 0, \quad h_1(t) - h_2(t) \rightarrow 0. \tag{133}$$

The theorem is an active-domain theorem, not global stability on all phase space: The active-domain assumption is essential:

$$x(0) \in \mathcal{D}_A. \tag{134}$$

Closed limit:

$$\zeta_0 = 0 \implies S_q = 0, \quad I_{\text{gate}} = 0, \quad \dot{\lambda} = -\dot{\sigma}, \quad \sigma + \lambda = \sigma_0 + \lambda_0. \tag{135}$$

Open-Gate theorem summary: Open-Gate adds only bounded S_q in $\dot{q}$; it strengthens walls but preserves the dissipative core.